

BSD Conjecture: The "Density of Points" Derived from Mathematical Determination

[Original Question]

"Elliptic curve" E The rank of the group of rational points above is, E of L functions $s = 1$ Prove that the rank of the group of rational points on an elliptic curve E is equal to the order of the zeros of its L -function at $s=1$.

This question asks why discrete information, such as the algebraic "number of rational points (rank)," coincides with continuous information, such as the analytical "behavior of the function (L-function)."

1. Intermediate development of synchronization: Points as accumulation of prescribed elements

Step A: Rational points are

not random grains of sand scattered in space, but rather "nodes" where underlying mathematical rules intersect.

$$\text{Rational Points} = \sum (\text{Consistent Intersections})$$

Rational points are not random; they are the "nodes" where mathematical definitions intersect.

Step B: The L-function, a "description of external waves"

The L-function is like a wave function that describes the "specified density" of the entire space. $s = 1$ The order of the zeros in a given point indicates the "predetermined degree of overlap" at that location.

$$\text{Order of Zero} \propto \text{Density of Definition}$$

The order of the zero represents the density of the underlying mathematical rules.

Step C: The number of intersections (points) of the necessary coincidence (synchronization)

provisions and the density of the provisions (zeros of the function) coincide as a necessary consequence of the results leaking out of the provisions.

Once the blueprint (rules) is decided, the number of "pillars (points)" to be erected is practically predetermined from the start.

$$\text{Rank}(E(\mathbb{Q})) = \text{ord}_{s=1} L(E, s)$$

The rank and the order must align; once the rules are set, the number of resulting points is already inherent within the system.

Critique of modern mathematics: The fallacy of separating discrete and continuous functions

The fallacy of modern mathematics lies in **treating "enumeration (algebra)" and "the variation of functions (analysis)" as separate universes and struggling to bridge the gap between them .**

- **They overlook the absence of inconsistency:** they are surprised, asking, "Why do things with such different properties coincide?" But in a mathematical space without inconsistency, the two are simply measuring the same "definition" using different methods.
- **The Labyrinth of "Measurement Methods":** Becoming so engrossed in the complex calculations of zeta functions and L-functions that we fail to notice the fact that the simple coercive force of "maintaining the ratio" is the underlying driving force.

The reason the BSD conjecture remains unsolved is that modern mathematics has forgotten the unity of **"points being defined shadows and functions being defined light,"** and is trying to analyze shadows and light separately.

Modern math fails by separating "counting" from "analysis," not realizing they are two ways of measuring the same underlying consistency.